

Lyapunov-Stable Limits for Dynamic Spectral-Entropy Regularization

thesis-IV working notes (HyMeKo)

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Abstract

We derive a Lyapunov-stable bound on the regularization weight λ in dynamic spectral-entropy regularization. Treating task loss as a Lyapunov function, the descent condition $\dot{V} < 0$ yields a closed-form upper bound $\lambda_{\max}(t)$ that depends on the inner product $\langle \nabla \mathcal{L}_{\text{task}}, \nabla \mathcal{L}_{\text{reg}} \rangle$. We propose a practical estimator using the spectral-trajectory KL divergence as a one-step observable proxy, and show the resulting schedule $\lambda_{\text{eff}}(t) = \lambda_0 \exp(-\eta \text{KL}_t)$ satisfies the descent condition when an empirical bound on the inner product holds. The schedule generalises three previously-independent arms (`kl_trajectory`, `entropy_adaptive`, `entropy_target`) into a single closed-loop controller, and links the empirical variance-reduction signature to a Lyapunov attractor argument.

1 Setting

Let $\theta \in \mathbb{R}^P$ denote the parameters of a neural network with layers $\{W_l\}_{l=1}^L$ and let $\mathcal{L}_{\text{task}}(\theta)$ be the task loss (cross-entropy in our experiments). Define the regulariser $\mathcal{L}_{\text{reg}}(\theta)$ and the total loss

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{task}}(\theta) + \lambda \mathcal{L}_{\text{reg}}(\theta).$$

At every regulariser-update step (every N batches), the regulariser is recomputed from the current θ . The optimizer takes a gradient step

$$\theta_{t+1} = \theta_t - \eta_{\text{lr}} g_t, \quad g_t = \nabla \mathcal{L}_{\text{task}}(\theta_t) + \lambda(t) \nabla \mathcal{L}_{\text{reg}}(\theta_t).$$

We treat $\lambda(t)$ as the controller’s free variable and ask: *what schedule $\lambda(t)$ guarantees that the task loss decreases?*

2 Lyapunov function and descent condition

Definition 1 (Lyapunov candidate). *Take $V(\theta) := \mathcal{L}_{\text{task}}(\theta)$. Then V is bounded below ($V \geq 0$ for cross-entropy with non-degenerate labels) and locally quadratic at any minimum.*

For the discretised system the first-order change in V along one optimizer step is

$$\Delta V_t := V(\theta_{t+1}) - V(\theta_t) = -\eta_{\text{lr}} \langle \nabla V, g_t \rangle + O(\eta_{\text{lr}}^2). \quad (1)$$

Substituting $g_t = \nabla \mathcal{L}_{\text{task}} + \lambda \nabla \mathcal{L}_{\text{reg}}$ and using $\nabla V = \nabla \mathcal{L}_{\text{task}}$:

$$\Delta V_t = -\eta_{\text{lr}} \left(\|\nabla \mathcal{L}_{\text{task}}\|^2 + \lambda \langle \nabla \mathcal{L}_{\text{task}}, \nabla \mathcal{L}_{\text{reg}} \rangle \right) + O(\eta_{\text{lr}}^2). \quad (2)$$

Proposition 1 (Descent condition). *For sufficiently small step size η_{lr} , $\Delta V_t < 0$ if and only if*

$$\|\nabla \mathcal{L}_{\text{task}}\|^2 + \lambda \langle \nabla \mathcal{L}_{\text{task}}, \nabla \mathcal{L}_{\text{reg}} \rangle > 0. \quad (3)$$

The proof is immediate from (??).

3 Lyapunov-safe upper bound on λ

Two regimes follow from (??):

Regime A: aligned gradients ($\langle \nabla \mathcal{L}_{\text{task}}, \nabla \mathcal{L}_{\text{reg}} \rangle \geq 0$). The descent condition is satisfied for *every* $\lambda \geq 0$. Practically: when the regulariser’s preferred direction also reduces task loss, any regulariser strength is safe. *Consequence*: we can grow λ aggressively in this regime.

Regime B: opposed gradients ($\langle \nabla \mathcal{L}_{\text{task}}, \nabla \mathcal{L}_{\text{reg}} \rangle < 0$). Solving (??) for λ :

$$\lambda < \lambda_{\max}(t) := \frac{\|\nabla \mathcal{L}_{\text{task}}\|^2}{|\langle \nabla \mathcal{L}_{\text{task}}, \nabla \mathcal{L}_{\text{reg}} \rangle|}. \quad (4)$$

This is the **Lyapunov-safe bound**. Beyond $\lambda_{\max}(t)$, the regulariser overpowers the task gradient and the system fails to descend on $\mathcal{L}_{\text{task}}$.

Theorem 1 (Stability under Lyapunov schedule). *If $\lambda(t) \leq \lambda_{\max}(t)$ for all t , the sequence $V(\theta_t)$ is monotonically non-increasing for sufficiently small η_{lr} . If additionally $\mathcal{L}_{\text{task}}$ is bounded below, the sequence converges and the optimizer reaches a stationary point of the combined loss.*

4 Practical estimator: KL trajectory as inner-product proxy

The exact $\langle \nabla \mathcal{L}_{\text{task}}, \nabla \mathcal{L}_{\text{reg}} \rangle$ requires backward passes for both gradients separately — prohibitive at every regulariser update. We introduce a cheaper observable.

Proposition 2 (KL as inner-product proxy). *Let p_t denote the (normalised) eigenvalue distribution of the network’s spectral adjacency at step t , and let $\text{KL}_t := D_{\text{KL}}(p_{t-1} \| p_t)$. Under regularisation acting on $\mathcal{L}_{\text{reg}} = H(p_t)$, the magnitude $|\text{KL}_t|$ is monotonically related to the magnitude $|\langle \nabla \mathcal{L}_{\text{task}}, \nabla \mathcal{L}_{\text{reg}} \rangle|$ during a single optimizer step, in the sense that small inner-product implies small spectrum motion (small KL_t) and vice versa.*

Sketch. A single gradient step of length η_{lr} moves the spectrum by an amount proportional to $\eta_{\text{lr}} \cdot \|\nabla \mathcal{L}_{\text{reg}}\|$ when the task and regulariser gradients are aligned, and by $\eta_{\text{lr}} \cdot \|\nabla \mathcal{L}_{\text{reg}} - \alpha \nabla \mathcal{L}_{\text{task}}\|$ for some α when they are opposed. The KL between consecutive spectra is a quadratic functional of this motion to leading order. The exact monotonicity requires Lipschitz bounds on H and is empirical in our setting; we treat the relationship as a working proxy. $\square$

Schedule from the proxy. Let $\eta > 0$ be a sensitivity parameter. The schedule

$$\lambda_{\text{eff}}(t) = \lambda_0 \cdot \exp(-\eta \text{KL}_t) \quad (5)$$

contracts λ when the spectrum moves rapidly (large KL_t) and recovers λ when the spectrum stabilises (small KL_t). With clamping $\lambda_{\text{eff}} \in [\lambda_{\min}, \lambda_{\max}^{\text{clamp}}]$, the controller is well-defined and bounded.

Corollary 1 (Schedule satisfies descent under bounded inner-product). *Suppose there exist constants $C_1, C_2 > 0$ such that*

$$|\langle \nabla \mathcal{L}_{\text{task}}, \nabla \mathcal{L}_{\text{reg}} \rangle| \leq C_1 \|\nabla \mathcal{L}_{\text{task}}\| \cdot \|\nabla \mathcal{L}_{\text{reg}}\|, \quad \text{KL}_t \geq C_2 \cdot |\langle \nabla \mathcal{L}_{\text{task}}, \nabla \mathcal{L}_{\text{reg}} \rangle|$$

during the descent regime. Then the schedule (??) satisfies $\lambda_{\text{eff}}(t) \leq \lambda_{\max}(t)$ for some $\eta > 0$ depending on $C_1, C_2, \|\nabla \mathcal{L}_{\text{reg}}\|$.

The first inequality is Cauchy-Schwarz; the second is the empirical content of Proposition ??.

5 The unified entropy_lyapunov regulariser

The schedule (??) unifies three previously-independent regulariser arms in our codebase:

- `kl_trajectory` computed KL_t but used it as a *standalone* penalty $\mathcal{L}_{\text{reg}} = \lambda \cdot \text{KL}_t$.

- **entropy_adaptive** multiplied λ by $\exp(\eta \ddot{H})$, using the entropy second derivative rather than KL.
- **entropy_target** introduced a target H^* but with a fixed λ .

The unified form is

$$\mathcal{L}_{\text{reg}}^{\text{lyap}}(t) = \lambda_{\text{eff}}(t) \cdot \left(\lambda_A (H_{\text{norm}} - H^*)^2 + \lambda_B H_{\text{norm}} \right), \quad \lambda_{\text{eff}}(t) = \lambda_0 \exp(-\eta \text{KL}_t). \quad (6)$$

This single arm provides:

- **Per-term weighting** (λ_A, λ_B) for multi-objective control between attractor and pull components.
- **Closed-loop feedback** ($\exp(-\eta \text{KL}_t)$) for descent stability.
- **Target attractor** (H^*) for IB-style compression.

It is implemented in `run_benchmark.py` as the `entropy_lyapunov` arm and tested as phase 10 on the 4-dataset stress matrix.

6 Variance-reduction connection

The empirical pattern that has held across ~ 108 experiments is: *the regulariser reduces seed-to-seed variance even when mean accuracy is unchanged*. The Lyapunov framework gives a candidate explanation.

Proposition 3 (Lyapunov attractor reduces seed variance). *Fix the random seed of weight initialisation; let $\theta_0(s)$ denote the initial parameters for seed s . Under a Lyapunov-safe schedule with $\lambda(t) \leq \lambda_{\text{max}}(t)$, the trajectories $\{\theta_t(s)\}$ are contracted toward a common attractor set $\mathcal{A} = \{\theta : \nabla \mathcal{L}_{\text{task}}(\theta) + \lambda \nabla \mathcal{L}_{\text{reg}}(\theta) = 0\}$. The attractor is parameterised by the regulariser; for spectral-entropy regularisers $\mathcal{A}$ is restricted to weight configurations with H_{norm} near H^* (or near 0 in the unnormalised form). Consequence: different initialisations converge to spectrally-similar models, reducing the seed-induced variance in final-epoch accuracy.*

This is informal but matches the observed effect. A rigorous version requires Polyak-Łojasiewicz-style bounds on the combined loss landscape; we leave it to follow-up work.

7 Failure modes the Lyapunov framing predicts

The schedule (??) predicts two failure modes consistent with our experimental data:

Failure mode 1: H^* degenerate. When H^* is set such that the regulariser’s attractor pulls in the direction of *increasing* task loss (Regime B), $\lambda_{\text{max}}(t)$ becomes finite and small. With λ_0 fixed and η insufficient to contract λ_{eff} below λ_{max} , the system fails to descend. *Empirical match:* $H^* = 1.0$ on spirals collapses from +0.688 to +0.040, exactly the expected pattern when the entropy attractor opposes the task gradient.

Failure mode 2: insufficient η . When the KL signal is small (smooth landscapes, e.g. MNIST plain MLP) the modulator $\exp(-\eta \text{KL}_t) \approx 1$ regardless of η , and the controller reduces to a constant- λ regulariser. *Empirical match:* **entropy_adaptive** (which uses entropy acceleration, an analogous signal) was task-dependent — it dominated on spirals (jagged landscape, large $\ddot{H}$) but was mid-pack on MNIST (smooth landscape).

8 Open questions

1. **Tight estimator.** Proposition ?? is empirical. A formal monotonicity result relating KL_t to the gradient inner product would give the Lyapunov bound a verifiable certificate.
2. **Adam-like normalisation.** The third schedule we sketched ($\lambda \propto m_t/\sqrt{v_t}$) is theoretically cleaner — it normalises by the running variance of the regulariser signal. We have not yet implemented or tested it.
3. **Constraint on λ_A, λ_B .** The per-term weights are currently independent. A principled coupling (e.g. $\lambda_A + \lambda_B = 1$) would reduce the hyperparameter dimension and enable a Pareto frontier analysis.

9 Implementation

The `entropy_lyapunov` arm in `run_benchmark.py` computes

- H_{norm} from the model’s spectral adjacency eigenvalue distribution;
- KL_t from `prev_eigvals` and the current eigenvals (already computed for `kl_trajectory`);
- λ_{eff} as $\exp(-\eta \text{KL}_t)$, clamped to $[0.1, 10]$;
- $\mathcal{L}_{\text{reg}}^{\text{lyap}}$ as in equation (??).

CLI flags: `--lyapunov-eta`, `--lam-a`, `--lam-b`, `--target-entropy`. Defaults $\eta=5.0$, $\lambda_A=\lambda_B=1.0$, $H^*=0.5$.

Phase 10 (`run_overnight_views_ph10.sh`) runs the arm on spirals, circles, MNIST plain MLP, CapsMLP MNIST.